# Integration & Applications

## Section 1: Indefinite Integrals & Power Rule

**1** Evaluate the indefinite integral: $\int (6x^2 - 4x + 3)\,dx$.

**a.** $2x^3 - 2x^2 + 3x + C$

**b.** $3x^3 - 2x^2 + 3x + C$

**c.** $2x^3 - 4x^2 + 3x + C$

**d.** $6x^3 - 2x^2 + 3x + C$

**2** Evaluate $\int 8x^3\,dx$.

**a.** $2x^4 + C$

**b.** $4x^4 + C$

**c.** $8x^4 + C$

**d.** $x^4 + C$

**3** What is $\int \frac{1}{x^2}\,dx$?

**a.** $-\frac{1}{x} + C$

**b.** $\frac{1}{x} + C$

**c.** $-\frac{2}{x^3} + C$

**d.** $\ln(x^2) + C$

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## Section 2: Definite Integrals

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**4** Evaluate the definite integral: $\int_0^3 2x\,dx$.

**a.** 9

**b.** 6

**c.** 18

**d.** 12

**5** Evaluate $\int_1^2 (3x^2 + 1)\,dx$.

**a.** 8

**b.** 7

**c.** 9

**d.** 10

**6** What is the value of $\int_{-2}^2 x^3\,dx$?

**a.** 0

**b.** 8

**c.** 16

**d.** $-8$

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## Section 3: Area Under Curves

**7** What is the area of the region bounded by $y = x^2$, the $x$-axis, and the lines $x = 0$ and $x = 3$?

**a.** 9

**b.** 6

**c.** 12

**d.** 27

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**8** What is the area of the region under the line $y = 4$ from $x = 1$ to $x = 5$?

**a.** 16

**b.** 20

**c.** 12

**d.** 8

**9** Find the area between the curve $y = 6x - x^2$ and the $x$-axis from $x = 0$ to its root $x = 6$.

**a.** 36

**b.** 18

**c.** 72

**d.** 54

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## Section 4: Fundamental Theorem of Calculus

**10** If $F(x) = \int_1^x (3t^2 - 2)\,dt$, what is $F'(x)$?

**a.** $3x^2 - 2$

**b.** $x^3 - 2x$

**c.** $6x$

**d.** $3x^2$

**11** If $\int_1^4 f(x)\,dx = 7$ and $\int_2^4 f(x)\,dx = 3$, what is $\int_1^2 f(x)\,dx$?

**a.** 4

**b.** 10

**c.** 2

**d.** 5

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**12** What is the value of $\int_5^5 (x^4 + 3x + 1)\,dx$?

**a.** 0

**b.** 1

**c.** 5

**d.** Undefined

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